

UNIVERSITÄT
LEIPZIG

International Physics Studies Program

Advanced Physics Laboratory

Pulsed ^{1}H NMR in Liquids

Calibration, Relaxation, Free-Induction Decay, Echo Measurements, and Chemical-Shift Uncertainty

Group: O4

Date: 14 April 2026

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Abstract

This report presents a pulsed ^{1}H NMR investigation of liquid samples using the MARAN Ultra spectrometer and the RiNMR software environment. The 90° and 180° pulse lengths were calibrated to $P90 = 2.85\,\mu s$ and $P180 = 5.70\,\mu s$, in agreement with the expected factor-of-two relation. Longitudinal relaxation times obtained by inversion recovery were $T_1 = 332.522$ ms for water, 6.035 ms for sample #1, 106.064 ms for sample #5, and 51.720 ms for oil, showing clear sample-dependent relaxation behaviour. The Fourier-transformed water FID gave a linewidth of approximately 529.7 Hz, corresponding to an effective transverse decay time of $T_2 \approx 601\,\mu s$.. The frequency-axis uncertainty of the chemical-shift measurement was estimated as $\sigma_\delta \approx 38.92$ ppm, with a maximum observed deviation of 59.05 ppm, indicating that the setup is not suitable for precise high-resolution chemical-shift determination. Echo-based measurements yielded $T_2 = (65.82 \pm 0.60)$ ms from the Hahn-echo fit for water, while CPMG gave reproducible values of 70.820 ms and 70.568 ms for water and a much larger value of 547.135 ms for sample #5. Overall, the measured relaxation times and pulse-calibration results are physically reasonable and consistent with the expected behaviour of liquid-state NMR.

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1. Introduction

Nuclear magnetic resonance (NMR) is one of the most important experimental methods for probing the microscopic properties of matter. It provides access to the magnetic behavior of atomic nuclei in an external magnetic field and makes it possible to study both structural and dynamical properties of materials. Because the resonance condition depends sensitively on the local magnetic environment of the nuclei, NMR has become a central tool in physics, chemistry, materials science, and medical imaging. In liquids in particular, the rapid molecular motion averages out many anisotropic interactions, producing comparatively narrow resonance lines and making the interpretation of the measured signals significantly more straightforward than in solids.

In pulsed NMR, the nuclear spin system is first brought out of equilibrium by short radio-frequency pulses and is then observed during its return toward equilibrium. This makes it possible to extract characteristic quantities such as the longitudinal relaxation time T_1, the transverse relaxation behavior, the free induction decay (FID), and the response of the spin system to refocusing sequences such as Hahn echo and Carr–Purcell–Meiboom–Gill (CPMG). These observables are directly linked to physically relevant processes including energy exchange with the surroundings, dephasing of transverse magnetization, magnetic-field inhomogeneity, and molecular motion. The theoretical lab manual used for this experiment provides the physical foundation for these effects, including the resonance condition, the action of $\pi/2$ and π pulses, relaxation processes, linewidth formation, chemical shift, and spin-echo sequences [1].

The present experiment was carried out on the MARAN Ultra spectrometer using the RiNMR software environment. In contrast to a high-resolution variable-field spectrometer, this setup is designed for instructional pulsed ^{1}H NMR experiments at fixed operating conditions, which makes it especially suitable for studying the fundamental signal formation and pulse-sequence concepts in a direct and transparent way. The practical workflow of the experiment followed the RiNMR guide used during the lab session [2]. In this guide, the measurements were organized into a sequence of tasks including calibration of the 90° and 180° pulse lengths, determination of T_1 by inversion recovery, acquisition and Fourier transformation of an FID signal, estimation of the uncertainty in chemical-shift measurements by controlled offset variation, and characterization of transverse relaxation using Hahn echo and CPMG pulse sequences.

Particular attention is given to the relation between pulse calibration and coherent spin manipulation, between inversion recovery and longitudinal relaxation, between FID decay and spectral linewidth, and between spin-echo sequences and the suppression of dephasing caused by magnetic-field inhomogeneity. In addition, the modified chemical-shift task is treated as an experimental estimate of measurement uncertainty rather than as a conventional high-resolution chemical-shift analysis.

Overall, the experiment serves as an integrated introduction to pulsed NMR in liquids. It illustrates how experimentally observed time-domain signals arise from controlled manipulations of nuclear magnetization, how these signals are transformed into frequency-domain information, and how relaxation and refocusing methods can be used to extract characteristic material parameters. The measurements thus provide a compact but physically rich demonstration of the central ideas underlying modern NMR spectroscopy.

2. General Theoretical Foundations

2.1 Basic Principle of Nuclear Magnetic Resonance

Nuclear magnetic resonance arises from the interaction between the magnetic moments of atomic nuclei and an externally applied magnetic field. Only nuclei with non-zero spin possess a magnetic moment and can therefore be observed by NMR. In the present experiment, the relevant nucleus is the proton, since the spectrometer is operated as a fixed-frequency ^{1}H NMR instrument for liquids. When a static magnetic field B_0 is applied, the previously degenerate nuclear spin states split into discrete Zeeman levels. The resulting population difference between these levels is small, but it is sufficient to produce a measurable macroscopic magnetization along the direction of the external field.

A useful feature of liquid-state NMR is that rapid molecular motion averages out many orientation-dependent interactions. As a result, the spectra are much simpler and narrower than in the solid state, and the measured signals can be interpreted largely in terms of isotropic averages. This is one of the main reasons why pulsed ^{1}H NMR in liquids is especially well suited for studying the basic principles of resonance, relaxation, and pulse-sequence-based signal formation.

2.2 Larmor Precession and Resonance Condition

In a magnetic field, the nuclear magnetic moments do not remain static but precess around the field direction. This motion is called Larmor precession, and its angular frequency is given by

$$\omega_L = \gamma B_0, \tag{1}$$

or, equivalently,

$$\nu_L = \frac{\gamma}{2\pi} B_0, \tag{2}$$

where γ is the gyromagnetic ratio of the nucleus. These relations define the resonance condition of the experiment: energy can be transferred efficiently to the spin system only when the applied radio-frequency field matches the Larmor frequency.

For the present experiment, this point is not only conceptually important but also experimentally visible. In RiNMR, the frequency offset $O1$ is adjusted until the measured signal is brought into resonance. If the offset is not chosen correctly, the signal shape changes noticeably in both the time and frequency domains. The need for repeated retuning during the experiment reflects the practical fact that the resonance condition is sensitive to instrumental drift.

2.3 Radio-Frequency Excitation and Pulse Lengths

At thermal equilibrium, the macroscopic magnetization points along the static field axis and is therefore not directly observable in the receiver coil. To generate a measurable signal, the magnetization must be tipped away from this axis by a radio-frequency pulse. In the rotating-frame picture, the RF field acts as an effective transverse magnetic field and rotates the magnetization by a defined angle. The flip angle θ is determined by the pulse duration t_p and the RF field strength B_1:

$$\theta = \gamma B_1 t_p. \tag{3}$$

Two pulse lengths are of central importance in this report. A 90° pulse rotates the equilibrium magnetization from the longitudinal direction into the transverse plane. A 180° pulse inverts the magnetization or, depending on the sequence, reverses the phase evolution of dephased spins. Since all later measurements depend on well-defined spin manipulations, the determination of the 90° and 180° pulse lengths is a necessary first step.

After a 90° pulse, the transverse magnetization induces a voltage in the detection coil. This time-dependent signal is the free induction decay. In practice, however, the signal cannot be observed immediately after the pulse because the spectrometer requires a short recovery time. This receiver dead time is especially relevant for rapidly decaying signals and is one reason why spin-echo techniques are so useful in pulsed NMR.

2.4 Longitudinal and Transverse Relaxation

Once the spin system has been perturbed by an RF pulse, it returns to equilibrium through relaxation processes. Two characteristic time scales are relevant here. The first is the longitudinal relaxation time T_1, which describes how quickly the magnetization component parallel to the static field recovers. Physically, this process is associated with energy exchange between the spin system and its surroundings, often referred to as the lattice. The second is the transverse relaxation time T_2, which describes the decay of phase coherence in the plane perpendicular to the static field.

These two relaxation processes affect different types of measurements. Longitudinal relaxation determines how quickly the system can recover between successive scans and forms the basis of the inversion-recovery experiment used to determine T_1. Transverse relaxation, by contrast, governs the decay of the observable transverse signal. In an ideal homogeneous field, this decay reflects intrinsic dephasing processes within the spin system. In a real spectrometer, however, field inhomogeneities lead to additional dephasing and therefore to an apparent signal decay that can be faster than the true T_2, denoted by T_2^*.

This distinction is particularly important for the present experiment. The free induction decay contains contributions both from genuine transverse relaxation and from static field inhomogeneity. For that reason, the decay observed directly after a 90° pulse does not necessarily represent the true transverse relaxation time. Echo-based sequences such as Hahn echo and CPMG are introduced later in the report precisely because they suppress part of this inhomogeneity-induced dephasing and therefore provide more reliable access to the intrinsic transverse relaxation behaviour.

3. Experimental Setup and Methods

3.1 Instrument and Software

The measurements were carried out on a MARAN Ultra pulsed NMR spectrometer using the RiNMR software environment. The experiment was performed as a liquid-state ^{1}H NMR study under fixed operating conditions, with the spectrometer frequency set in the acquisition parameters and the resonance condition adjusted through the offset parameter $O1$. The instrument was operated in pulsed mode, so that the nuclear magnetization was excited by short radio-frequency pulses and the resulting time-dependent signal was recorded digitally for further analysis.

RiNMR provides separate *Acquisition* and *Processing* modes. In the acquisition mode, the relevant pulse sequences were loaded and the experimental parameters were adjusted. In the processing mode, the recorded signals could be inspected and analyzed further. The pulse timing scheme of a loaded sequence was visualized using the pulse viewer, while live measurements for parameter adjustment were monitored with the `GS1` command. For the actual data acquisition, the `GO` command was used. During all measurements, care was taken to avoid clipping of the signal by choosing an appropriate receiver gain.

Since the resonance offset was found to drift during the experiment due to heating of the spectrometer, the automatic frequency adjustment routine (`Auto 01`) was performed regularly before most measurements. This ensured that the spectrometer remained close to resonance throughout the experiment. In addition, the relaxation delay RD and the receiver gain RG were adapted for the individual samples and pulse sequences in order to obtain stable signals of suitable intensity.

3.2 Samples

The experiment was carried out on several liquid samples relevant to the selected tasks. According to the practical guide, the required measurements comprised two different water samples, n-butanol (sample #5), and oil for the inversion-recovery measurements. The free induction decay was measured for a water sample, whereas the Hahn echo and CPMG experiments were performed for one water sample and for n-butanol. Optional measurements on methylbenzene and HMDSO mentioned in the older manual were not part of the required experimental program considered here.

The older manual lists several water-based and organic samples, including tap water, $NiSO_4$-doped water, n-butanol, methylbenzene, and HMDSO. In the present report, however, only those samples that were actually required by the updated RiNMR guide are considered in the evaluation. This distinction is important because the theoretical manual describes a broader task set than the one that was finally performed in the laboratory.

3.3 General Measurement Procedure

At the beginning of the experiment, a test measurement was carried out using a water sample in order to become familiar with the instrument and the software. For this purpose, the FID pulse sequence (`fid.exe`) was loaded and the default acquisition parameters were checked. The resonance condition was then adjusted through the offset parameter $O1$, and the response of the FID signal to changes in this parameter was observed. This initial procedure served both as a practical introduction to RiNMR and as a first check that the spectrometer was properly tuned.

After this initial setup, the experiment followed the sequence of tasks prescribed in the practical guide. First, the 90° and 180° pulse lengths were adjusted using a water sample. This was done both by manually varying the pulse duration and by using the automatic pulse calibration routine. In the next step, the longitudinal relaxation time T_1 was determined by means of the inversion-recovery pulse program (`invrec.exe`) for the required set of samples. For each sample, suitable relaxation delays and lists of delay times were chosen so that the magnetization recovery could be recorded over the relevant time range.

The subsequent measurements focused on time-domain and echo-based signal analysis. A free induction decay was recorded for a water sample using the FID sequence, then Fourier transformed and phased in RiNMR. The resulting frequency-domain peak was exported and fitted externally in order to determine its full width at half maximum. To estimate the uncertainty of chemical-shift measurements, the same FID-based measurement was repeated at different offset values, and the resulting spectral shifts were compared. Hahn echo measurements were then performed with the sequence `Hahn.exe` by varying the pulse spacing τ and recording the corresponding echo amplitudes. Finally, CPMG measurements were carried out with `cpmg.exe`, again for a water sample and for n-butanol, in order to analyze the decay of a train of refocused echoes.

The diffusion experiment described in the older manual was not performed, since the required gradient coils were not installed in the setup used for the laboratory course. It is therefore not included in the present report.

4. Results and Discussion

4.1 Calibration of the 90° and 180° Pulse Lengths

4.1.1 Theoretical Background

The first step of the experiment consisted of determining the pulse lengths required to rotate the equilibrium magnetization by 90° and 180°. This calibration is essential because all subsequent pulse sequences assume that these rotations are carried out accurately. In pulsed NMR, a short radio-frequency pulse applied perpendicular to the static magnetic field B_0 tips the macroscopic magnetization away from its equilibrium direction. In the rotating-frame picture, the pulse acts as an effective transverse field B_1, and the resulting flip angle θ is given by [1]

$$\theta = \gamma B_1 t_p, \tag{4}$$

Where:

γ : gyromagnetic ratio,
t_p : pulse duration.

A 90° pulse rotates the equilibrium magnetization from the longitudinal direction into the transverse plane. This is the configuration that produces the largest directly observable NMR signal, since only the transverse component induces a voltage in the receiver coil. A 180° pulse, by contrast, inverts the magnetization and is required for refocusing and inversion-based pulse sequences used later in the experiment. For ideal excitation conditions, the corresponding pulse lengths satisfy [1]

$$\gamma B_1 \tau_{\pi/2} = \frac{\pi}{2}, \tag{5}$$

and

$$\gamma B_1 \tau_{\pi} = \pi. \tag{6}$$

It follows immediately that the 180° pulse length is ideally twice the 90° pulse length.

The calibration can be understood most easily by considering how the free induction decay depends on the created transverse magnetization. In an idealized picture, the observable transverse component after a pulse is proportional to the sine of the flip angle [2],

$$M_{xy} \propto \sin\theta. \tag{7}$$

This means that the signal amplitude increases as the pulse length approaches the 90° condition, reaches a maximum at $\theta = \pi/2$, and decreases again for longer pulses. A manual scan of the pulse duration therefore provides an experimental estimate of the 90° pulse length, while the position of the 180° condition follows at approximately twice that value.

In practice, the calibration is not perfectly ideal. One important limitation is the spatial inhomogeneity of the RF field across the sample volume. If the field strength B_1 is not exactly the same everywhere in the sample, different parts of the magnetization experience slightly different flip angles during the same pulse. This broadens the calibration maximum and makes the experimentally observed optimum less sharp than in the ideal case. For this reason, pulse calibration is usually more reliable when based on the overall signal maximum rather than on an attempt to infer the rotation angle from a single trace alone.

A second practical limitation is the dead time of the spectrometer. Immediately after the RF pulse, the receiver cannot detect the signal because of electronic recovery and ring-down effects. As a result, the earliest part of the free induction decay is not directly observable. Although this does not prevent pulse calibration, it explains why the measured FID begins only after a short delay and why very rapidly decaying signals are more difficult to assess directly. The calibration in this task therefore relies mainly on the overall signal amplitude observed after the pulse rather than on the exact form of the earliest part of the decay.

Task 1 is therefore best understood as a calibration of coherent spin manipulation. By varying the pulse duration and monitoring the resulting FID amplitude, the pulse lengths corresponding to 90° and 180° rotations can be identified. These values then form the basis for all later measurements, including inversion recovery, Hahn echo, and CPMG experiments.

4.1.2 Measurement Procedure

For the calibration of the 90° and 180° pulse lengths, a water sample was inserted into the probehead and the FID pulse sequence (`fid.exe`) was loaded in the acquisition mode of RiNMR. Before starting the pulse calibration itself, the spectrometer was brought close to resonance by adjusting the offset parameter $O1$, following the standard tuning procedure described in the practical guide. The live acquisition mode `GS1` was then used so that the response of the free induction decay could be monitored directly while changing the pulse length. Care was taken to keep the receiver gain at a value for which the signal was not clipped. [2]

The pulse parameter $P90$ was first set to 1.0 µs. The remaining relevant parameters were kept at

$$RD = 2\,000\,000\,\mu s, \qquad NS = 4, \qquad DS = 0.$$

The dependence of the FID amplitude on the pulse length was first examined qualitatively in live mode by varying the parameter $P90$. Instead of documenting the full scan over all intermediate pulse lengths, representative comparison traces were saved for one value below the optimum, one at the optimum, and one above the optimum. Specifically, traces were recorded for $P90 = 2.00\,\mu s$, $P90 = 2.85\,\mu s$, and $P90 = 4.00\,\mu s$. These traces were used to illustrate the expected trend that the signal amplitude increases toward the 90° condition and decreases again for longer pulses.

After this qualitative manual check, `GS1` was stopped, and the automatic calibration routine was executed. This routine returned the pulse lengths

$$P90 = 2.85\,\mu s$$

and

$$P180 = 5.70\,\mu s.$$

A screenshot of the calibration result was saved directly from the RiNMR window, as recommended in the guide. The value found for $P180$ is exactly twice the value of $P90$, which is consistent with the ideal relation between the two pulse lengths. The traces recorded at 2.00 µs, 2.85 µs, and 4.00 µs were then used for the comparison.

4.1.3 Results

To illustrate the dependence of the signal on pulse duration, representative FID traces were documented for three values of $P90$: 2.00 µs, 2.85 µs, and 4.00 µs as shown in fig. 1. The trace recorded at 2.85 µs, corresponding to the value determined by the automatic calibration, showed the strongest overall signal among the documented cases. By contrast, the traces recorded at 2.00 µs and 4.00 µs exhibited smaller amplitudes, consistent with excitation below and above the optimum 90° condition, respectively.

A direct comparison of the calibrated values further shows that

$$P180 = 2 \cdot P90,$$

since

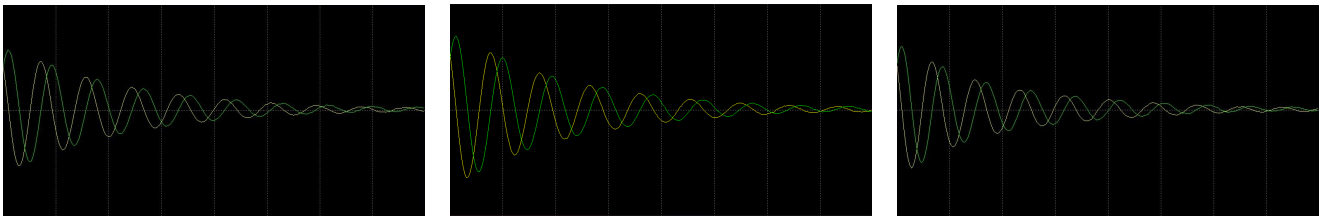
$$5.70\,\mu s = 2 \times 2.85\,\mu s.$$

Thus, the experimentally determined pulse lengths are consistent with the expected relation between the 90° and 180° pulses. The calibrated values were therefore used for all subsequent measurements in the experiment.

4.1.4 Discussion

The pulse-length calibration produced physically consistent results. The automatic routine yielded a 90° pulse length of 2.85 µs and a 180° pulse length of 5.70 µs, so that the expected ideal relation between the two pulse lengths was fulfilled:

$$P180 = 2 \cdot P90.$$



(a) $P90 = 2.00\,\mu s$ **(b)** $P90 = 2.85\,\mu s$ **(c)** $P90 = 4.00\,\mu s$

Figure 1: Representative free induction decay traces recorded during the pulse-length calibration. The middle panel corresponds to the automatically determined optimal 90° pulse length, $P90 = 2.85\,\mu s$, while the left and right panels show comparison measurements for shorter and longer pulse durations, respectively.

This agreement is an important internal consistency check, since the 180° pulse should ideally correspond to twice the duration of the 90° pulse when the RF field amplitude is kept constant. The result therefore indicates that the automatic calibration identified a plausible operating point for the subsequent measurements.

The comparison of the representative FID traces further supports this interpretation. Among the documented measurements, the signal recorded for $P90 = 2.85\,\mu s$ showed the strongest response, whereas the signals obtained for $P90 = 2.00\,\mu s$ and $P90 = 4.00\,\mu s$ were weaker. This behaviour is exactly what is expected from the dependence of the transverse magnetization on the flip angle: for pulse lengths below the optimum, the magnetization is not yet fully rotated into the transverse plane, while for longer pulse lengths the rotation exceeds the 90° condition and the observable transverse component decreases again. In this sense, the comparison measurements below and above the optimum provide qualitative confirmation of the calibration result.

It should be noted, however, that the full manual scan from 1 µs to 15 µs, as suggested in the practical guide [2], was not documented point by point in the present report. Instead, the trend was checked qualitatively during live observation and then illustrated using one representative value below the optimum and one above it. This is sufficient to demonstrate the expected signal behaviour, but it does not provide a full nutation-type curve of signal amplitude versus pulse length. The final calibrated values therefore rely primarily on the `Auto P90` routine rather than on a manual quantitative determination.

A further point of practical relevance is that the pulse calibration is influenced by non-ideal experimental conditions. The older theoretical manual notes in particular the role of RF-field inhomogeneity across the sample volume and the dead time of the spectrometer [1]. An inhomogeneous B_1 field means that not all parts of the sample experience exactly the same flip angle during a pulse, which broadens the optimum and makes the maximum signal less sharp than in an idealized model. In addition, the earliest part of the FID cannot be detected directly because of receiver recovery after the pulse. These effects do not invalidate the calibration, but they explain why the experimentally observed optimum is determined operationally from the overall signal strength rather than from a perfectly sharp theoretical criterion.

4.2 Determination of T_1 by Inversion Recovery

4.2.1 Theoretical Background

Here the determination of the longitudinal relaxation time T_1 by means of the inversion-recovery method is discussed. In contrast to the free induction decay, which reflects the evolution of transverse magnetization, this experiment probes how the magnetization component parallel to the static magnetic field returns to thermal equilibrium after it has been inverted. The manual explicitly defines this task as the determination of the longitudinal relaxation time using the $\pi-\tau-\pi/2$ pulse group [1].

Longitudinal relaxation describes the recovery of the equilibrium magnetization along the direction of the external magnetic field. Physically, this process is associated with energy exchange between the spin system and its surroundings and is therefore commonly referred to as spin-lattice relaxation. If the longitudinal magnetization is brought away from equilibrium, its recovery can be described in simplified form by

$$M_z(t) = M_0\left(1 - e^{-t/T_1}\right), \tag{8}$$

where M_0 denotes the equilibrium magnetization. The characteristic time constant of this recovery is T_1.

In practice, the longitudinal magnetization itself cannot be observed directly in the spectrometer. For this reason, the inversion-recovery sequence converts longitudinal recovery into a measurable signal. First, a π pulse inverts the equilibrium magnetization from the positive z direction to the negative one. After this, the system is allowed to recover for a variable delay time τ. Finally, a $\pi/2$ pulse is applied, which rotates the remaining longitudinal magnetization into the transverse plane, where it can be detected as an NMR signal. The sign and magnitude of this signal therefore reflect how far the recovery has proceeded during the delay time τ. After a π pulse the population difference is inverted, and the subsequent $\pi/2$ pulse converts the recovering longitudinal state into an observable signal [1].

For short delay times, the magnetization is still predominantly negative when the readout pulse is applied, so the measured signal is negative. With increasing τ, the longitudinal magnetization recovers toward equilibrium, and the detected signal increases. In the ideal inversion-recovery experiment, the signal follows

$$M(\tau) = M_{\mathrm{eq}} \left(1 - 2e^{-\tau/T_1}\right). \tag{9}$$

It explains the characteristic shape of the measured curve: the signal begins at a negative value, passes through zero, and approaches the positive equilibrium value for large delay times [1].

A particularly useful feature of the inversion-recovery curve is its zero crossing. Setting the measured magnetization equal to zero gives

$$0 = 1 - 2e^{-\tau_0/T_1}, \tag{10}$$

and therefore

$$\tau_0 = T_1 \ln 2. \tag{11}$$

Thus, the delay time at which the signal changes sign already provides a simple estimate of T_1. This relation is valid provided the pulse lengths are adjusted correctly and the RF field is sufficiently homogeneous.

This principle was implemented in RiNMR by using the inversion-recovery sequence `invrec.exe`, in which the delay parameter $D1$ corresponds to the recovery time τ. A list of different $D1$ values were measured, and the resulting signal values were fitted automatically to obtain T_1. It is also recommended to choose the relaxation delay RD long enough that repeated scans do not distort the recovery curve, typically such that $RD \geq 5T_1$. Although this belongs to the practical implementation rather than the underlying physics, it reflects the same central idea: the experiment is only meaningful if each measurement starts from a nearly recovered equilibrium state [1].

All in all, the purpose here was monitoring how quickly an inverted spin system returns to equilibrium. The inversion-recovery sequence is especially suitable for this purpose because it transforms a non-observable longitudinal process into a directly measurable signal amplitude, whose dependence on the delay time τ can be fitted quantitatively.

4.2.2 Measurement Procedure

For the determination of the longitudinal relaxation time T_1, the inversion-recovery pulse sequence was used in the acquisition mode of RiNMR. The corresponding pulse program `invrec.exe` was loaded, and the pulse timing was inspected with the pulse viewer using the `.TD` command. Before each measurement, the spectrometer was tuned close to resonance by checking the offset parameter $O1$, since the practical guide notes that the resonance offset can drift during the experiment because of spectrometer heating. The receiver gain was adjusted so that the signal intensity was clearly visible but not clipped [2].

The variable delay parameter of the inversion-recovery experiment was the quantity $D1$, which corresponds to the recovery time τ between the initial π pulse and the subsequent $\pi/2$ readout pulse. Following the practical guide, $D1$ was initially set to 1000 µs. The receiver gain RG was then adapted to the respective sample so that a suitable signal intensity was obtained. The guide gives typical starting values such as $RG = 2$ for the reference water sample, $RG = 8$ for sample #5, and $RG = 50$ for sample #1, depending on the sample volume and signal strength [2].

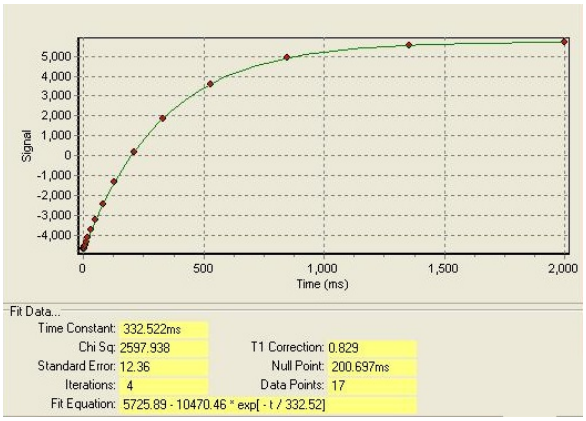
A suitable value for the relaxation delay RD was estimated before the full measurement series was started. This was done according to one of the two procedures recommended in the practical guide: either a preliminary measurement with approximately 5–8 different $D1$ values was performed using $RD = 2$ s and checked against the condition $RD \geq 5T_1$, or RD was increased stepwise starting from 100 000 µs until the measured signal intensity no longer changed appreciably. In this way, the repetition delay was chosen long enough that each scan started from a nearly recovered equilibrium state.

After this preparation, the RiNMR script `T1.ris` was used for the actual acquisition. For this purpose, a list file named `T1_list` was opened or created, containing the selected $D1$ values in units of microseconds, entered one below another. The delay values were distributed approximately homogeneously or logarithmically over the relevant time range, beginning at 1000 µs and extending up to a maximum delay of the order of the estimated RD. This ensured that the negative part, the zero crossing, and the saturation region of the inversion-recovery curve were all covered sufficiently for a reliable fit.

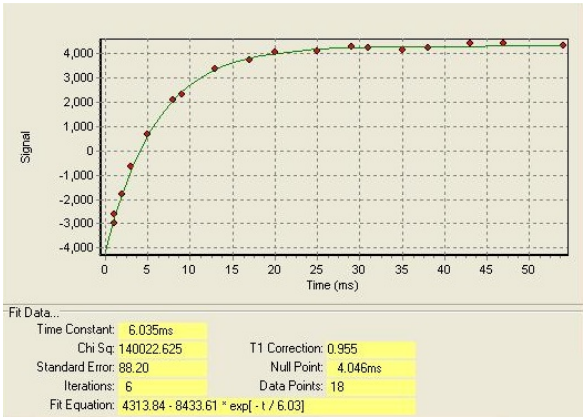
The first inversion-recovery measurement was performed for a water sample. After completion of the measurement, RiNMR displayed the fitted recovery curve together with the extracted value of T_1. The same procedure was then repeated for the other required samples of the updated practical program. According to the RiNMR guide, this comprised two different water samples, n-butanol (sample #5), and oil. For each sample, the values of RG, RD, and if necessary the $D1$ list were readjusted so that the fit could be performed under suitable signal conditions. The guide also notes that the signal intensity should not fall below about 100, otherwise the fit may become unreliable [2].

4.2.3 Results

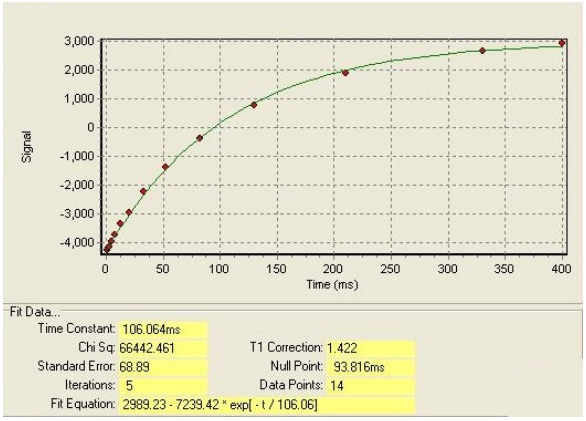
The inversion-recovery measurements yielded well-defined recovery curves for all investigated samples. In each case, the signal started at negative values for short delay times, crossed zero at an intermediate delay, and approached a positive saturation value for larger delays. This behaviour is characteristic of longitudinal magnetization recovery and is



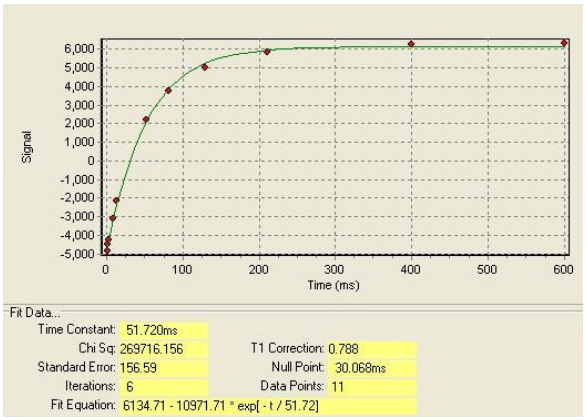
(a) Water



(b) Sample #1



(c) Sample #5



(d) Oil

Figure 2: Inversion-recovery fits obtained for the investigated samples.

consistent with the expected form of the inversion-recovery experiment.

The fitted longitudinal relaxation times obtained from the RiNMR evaluation are summarized in Table 1. The largest value was found for the water sample, while the smallest value was obtained for sample #1. The measurements for oil and sample #5 yielded intermediate relaxation times.

Table 1: Fitted longitudinal relaxation times obtained from the inversion-recovery measurements.

Sample	T_1 (ms)	Null point (ms)
Water	332.522	200.697
Sample #1	6.035	4.046
Sample #5	106.064	93.816
Oil	51.720	30.068

For the water sample, the fit resulted in

$$T_1 = 332.522\,\text{ms},$$

with a null point at

$$\tau_0 = 200.697\,\text{ms}.$$

This was the largest T_1 value among the measured samples and corresponds to the slowest recovery of longitudinal magnetization. The fitted curve approached its saturation value only on the timescale of several hundred milliseconds.

For sample #1, the recovery was much faster. The fit yielded $T_1 = 6.035\,\text{ms}$, with a null point at $\tau_0 = 4.046\,\text{ms}$. This was by far the smallest relaxation time measured, indicating that the longitudinal magnetization recovered almost completely within only a few milliseconds. For sample #5, the fitted result was $T_1 = 114.491\,\text{ms}$, with a null point at $\tau_0 = 95.173\,\text{ms}$. Its relaxation behaviour was therefore significantly slower than that of sample #1, but still clearly faster than that of the water sample. For the oil sample, the fit gave $T_1 = 51.720\,\text{ms}$, with a null point at $\tau_0 = 30.068\,\text{ms}$. This places the oil sample between sample #1 and sample #5 in terms of longitudinal relaxation time.

Overall, the measured values span more than one order of magnitude, from approximately 6 ms to 333 ms. The inversion-recovery method therefore clearly resolved substantial differences in the longitudinal relaxation behaviour of the investigated samples.

4.2.4 Discussion

The inversion-recovery measurements clearly resolved substantial differences in the longitudinal relaxation

behaviour of the investigated samples. The fitted values ranged from only 6.035 ms for sample #1 up to 332.522 ms for the water sample, with intermediate values of 51.720 ms for oil and 114.491 ms for sample #5. Thus, the experiment covered more than one order of magnitude in T_1, which shows that the inversion-recovery method was sufficiently sensitive to distinguish the samples clearly.

The measured water value, $T_1 = 332.522\,\text{ms}$, is in very good agreement with the practical guide, which states that a typical fitted value for the water sample is about 300 ms to 350 ms. This indicates that the tuning, pulse calibration, choice of delay times, and overall fitting procedure were all reasonable for this measurement. [2]

By contrast, sample #1 showed an extremely short relaxation time, $T_1 = 6.035\,\text{ms}$. This result is physically plausible if sample #1 is identified with the $NiSO_4$-containing water sample listed in the older manual. The theoretical manual explicitly notes that proton relaxation in water can be shortened strongly by interaction with paramagnetic ions such as Ni^{2+}, and that this reduction of T_1 into the millisecond range is in fact useful for rapid calibration measurements. The very small value obtained here is therefore not anomalous, but entirely consistent with the intended role of such doped water samples in the experiment. [1] [2]

Sample #5, which corresponds to n-butanol in the original sample list, yielded $T_1 = 114.491\,\text{ms}$. Its longitudinal relaxation is therefore considerably slower than that of the $NiSO_4$-doped water sample, but still distinctly faster than that of the water sample measured here. The oil sample, with $T_1 = 51.720\,\text{ms}$, falls between these two extremes. In qualitative terms, these results show that different molecular environments and different relaxation mechanisms lead to clearly distinguishable recovery times, even within the limited scope of a teaching experiment.

The null points obtained from the RiNMR fits are of the correct order of magnitude, but they do not coincide exactly with the ideal relation $\tau_0 = T_1 \ln 2$ for every sample. This is not surprising. The relation is derived for an ideal inversion-recovery experiment with perfectly adjusted pulse lengths and a homogeneous RF field, whereas the real experiment is influenced by practical limitations such as imperfect pulse calibration, RF-field inhomogeneity, limited signal-to-noise ratio, and baseline uncertainties. The theoretical manual itself states that the zero-crossing method is simple but presupposes well-adjusted pulses and a homogeneous RF field. The deviations observed here therefore reflect the non-ideal character of the real measurement rather than a failure of the method. [1]

A further limitation is that the reported values were taken directly from the RiNMR fits, so the discussion is based on the software output rather than on an independent refit of the raw data. This is acceptable for the purpose of the practical report, but it means that the quoted values should be understood as fit results within the RiNMR evaluation model. In particular, the displayed "T1 correction" factors indicate that the fitting routine is not based on a perfectly idealized recovery curve alone, which also helps explain why the fitted null points do not always reproduce the simplest analytical estimate exactly.

4.3 Free Induction Decay and Fourier Transformation

4.3.1 Theoretical Background

Data Loading: Four files were loaded for this analysis: two corresponding to the free induction decay (FID), a data file and its associated parameter file, and two corresponding to the fast Fourier transform (FFT), again consisting of a data file and a parameter file. The spectrometer frequency was then extracted from the parameter data.
from the parameter file using:

$$SF = 23.7006\ \text{MHz} \tag{12}$$

Time Domain Free Induction Decay (FID): The FID data consists of three columns: time (s), real intensity, and imaginary intensity. The time axis was converted to milliseconds for readability. Both the real and imaginary components were plotted, as the spectrometer uses quadrature detection to record a complex signal simultaneously in two channels that are 90° out of phase with each other. The FID appears as a decaying oscillation, immediately after the 90° pulse the magnetisation is at its maximum, and it decays over time due to transverse relaxation and field inhomogeneity.
Frequency Domain (Fourier-Transformed Spectrum): The Fourier transform converts the time-domain FID into a frequency-domain spectrum. After phasing to correct for lineshape distortions, the real part of the spectrum shows a symmetric peak centred near 0 MHz, corresponding to the water ^{1}H resonance.
Lorentzian Fit and FWHM Extraction: NMR peaks in solution are well described by a Lorentzian lineshape:

$$L(\nu) = \frac{A \cdot \gamma^2}{(\nu - \nu_0)^2 + \gamma^2} + \text{offset} \tag{13}$$

where ν_0 is the peak position, γ is the half-width at half maximum (so that FWHM = 2γ), and A is the amplitude. The fit was performed only on the central peak region (within ±0.04 MHz of the peak maximum) to avoid baseline noise affecting the result.
Calculating T_2: The transverse relaxation time T_2 is directly related to the linewidth. The full width at half maximum 1 / 2 is given by:

$$\delta\nu_{1/2} = \frac{1}{\pi T_2} \tag{14}$$

Rearranging:

$$T_2 = \frac{1}{\pi \cdot \delta\nu_{1/2}} \tag{15}$$

A broader peak corresponds to faster transverse relaxation and hence a shorter T_2.

4.3.2 Measurement Procedure

The FID signal of the selected water sample was measured using the standard FID pulse sequence on the spectrometer. After setting appropriate acquisition parameters based on the previously determined relaxation time, the system was tuned to obtain a clear and stable signal. The measurement was then carried out, and the time-domain FID signal was recorded and stored. This signal was subsequently Fourier transformed to obtain the frequency spectrum, followed by manual phase correction to achieve a symmetric peak shape. Finally, the processed data around the central peak was exported and fitted with a Lorentzian function to extract the full width at half maximum (FWHM).

4.3.3 Results

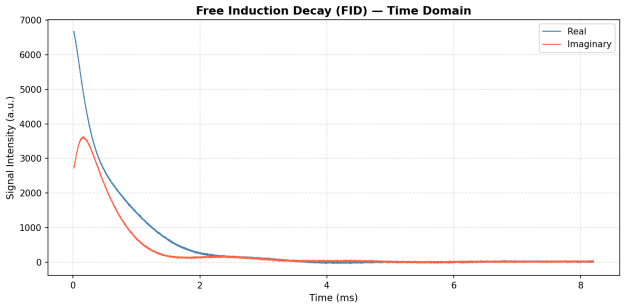


Figure 3: Time-domain FID signal of the water sample

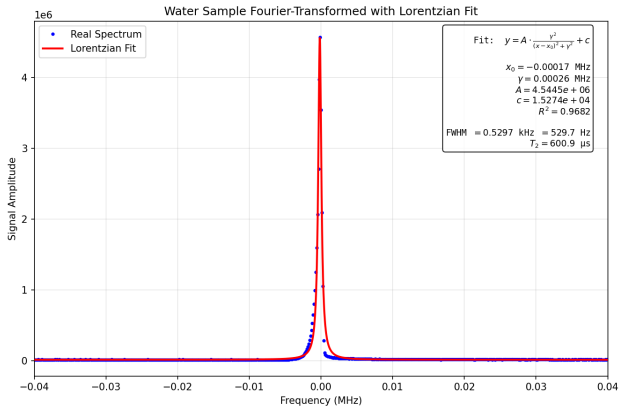


Figure 4: Fourier-transformed water spectrum with Lorentzian fit

Time-Domain FID Signal The free induction decay signal of the water sample is shown in Fig. 3. Both the real and imaginary channels are plotted, recorded simultaneously via quadrature detection. Immediately after the 90° pulse, the real channel reaches its maximum intensity of approximately 6,700 a.u. The imaginary channel peaks slightly later at around 3,600 a.u., reflecting the 90° phase offset between the two detection channels. Both signals decay smoothly toward zero within approximately 8 ms, consistent with exponential transverse relaxation. No oscillatory beating is visible, confirming that the carrier frequency was set very close to the water resonance frequency.

Frequency-Domain Spectrum and Lorentzian Fit The Fourier-transformed spectrum shows a single sharp peak centred near $\nu_0 \approx 0$ MHz, corresponding to the ^{1}H resonance of water. A Lorentzian fit was performed on the central peak region within ±0.04 MHz of the maximum. The fit parameters are summarised in Table 2.

Parameter	Value	Uncertainty
Peak position ν_0	−0.00017 MHz	±0.000005 MHz
Half-width γ	0.00026 MHz	—
FWHM = 2γ	0.000530 MHz	—
Amplitude A	4,544,500 a.u.	±91,000 a.u.
Offset c	15,274 a.u.	—
R^2	0.9682	—

Table 2: Fit parameters of the Lorentzian profile for the water ^{1}H spectrum.

The transverse relaxation time T_2 is obtained from the FWHM via equation (36):

$$T_2 = \frac{1}{\pi \cdot \mathrm{FWHM}_{\mathrm{Hz}}} = \frac{1}{\pi \cdot 529.7\ \mathrm{Hz}} \approx 6.01{\times}10^{-4}\ \mathrm{s} \approx 601\ \mu\mathrm{s} \tag{16}$$

The fit quality of $R^2 = 0.9682$ confirms a good agreement between the Lorentzian model and the measured spectrum over the fitted region.

4.3.4 Discussion

The FID signal shows the expected behaviour for a liquid sample: a smooth, exponentially decaying envelope with no visible oscillations or beating. The real channel begins at maximum intensity immediately after the 90° pulse, while the imaginary channel, being 90° out of phase, rises to its peak shortly after. Both channels decay to the noise level within approximately 8 ms, indicating that the transverse magnetisation is lost on this timescale.

The Lorentzian lineshape is expected for NMR signals in liquids because the transverse magnetisation decays exponentially over time, and the Fourier transform of an exponential decay is a Lorentzian function. The linewidth and T_2 are related by equation (36):

$$\delta\nu_{1/2} = \frac{1}{\pi T_2} \tag{17}$$

The measured FWHM of 529.7 Hz gives $T_2 \approx 601\ \mu$s. This value is physically consistent with the observed FID decay: a T_2 of 601 μs predicts that the signal should decay to $1/e$ of its initial value after approximately 0.6 ms, and indeed the FID envelope falls steeply in the first 1–2 ms before settling to the noise baseline by around 8 ms. The two observations are therefore mutually consistent.

The measured T_2 represents an effective relaxation time that includes contributions from both intrinsic transverse relaxation and residual magnetic field inhomogeneity across the sample. At this low field strength, inhomogeneity broadening is a significant contribution, meaning the true

intrinsic T_2 of water may be somewhat longer than 601 μs.

The peak position $\nu_0 \approx -0.00017$ MHz is very close to zero, indicating that the carrier frequency was set correctly and the offset from the water resonance is negligible. The fit quality ($R^2 = 0.9682$) is good, with small deviations in the wings of the peak attributable to noise and minor baseline imperfections, which are not captured by an ideal Lorentzian model.

4.4 Estimation of the Uncertainty in Chemical-Shift Measurements

4.4.1 Theoretical Background

What happens to the signal upon changing O1?: In the time domain (FID), changing O1 shifts the oscillation frequency of the signal, since O1 defines the reference frequency of the receiver — a higher O1 means the FID oscillates faster, and a lower O1 means it oscillates slower. In the frequency domain, the peak simply translates rigidly left or right by the applied offset (±10 kHz), with no change in lineshape or intensity.

Peak positions from Lorentzian fits:
The three spectra were fitted with a Lorentzian lineshape to extract precise peak positions. The −10 kHz and +10 kHz spectra are expected to shift by ±10 kHz relative to the reference, which in ppm units corresponds to

$$\pm \frac{10000}{SF_{\mathrm{Hz}}} \times 10^6 \text{ ppm}.$$

The small difference between the measured and expected positions is the measurement deviation.

Estimated uncertainty in chemical shift:
The deviations of the measured peak positions from their expected positions (printed as `deviation` in the output) represent the uncertainty in this type of chemical shift measurement. The mean of the two deviations gives the overall estimated uncertainty in ppm.

Comparison with typical chemical shifts:
Typical ^{1}H chemical shifts span 0–12 ppm, and ^{13}C shifts span 0–220 ppm. The estimated measurement uncertainty of a fraction of a ppm is therefore negligibly small compared to the range of chemically meaningful shifts, confirming that frequency-based chemical shift measurements on this spectrometer are highly reliable.

4.4.2 Measurement Procedure

A water sample was measured using the FID pulse sequence with $SI = 8192$ and `AUTO O1` applied to centre the carrier frequency. Three spectra were recorded: a reference spectrum, and two additional spectra with O1 shifted by +10 kHz and −10 kHz respectively, without reapplying `AUTO O1`. All three spectra were Fourier transformed, phased, and exported for analysis.

4.4.3 Results

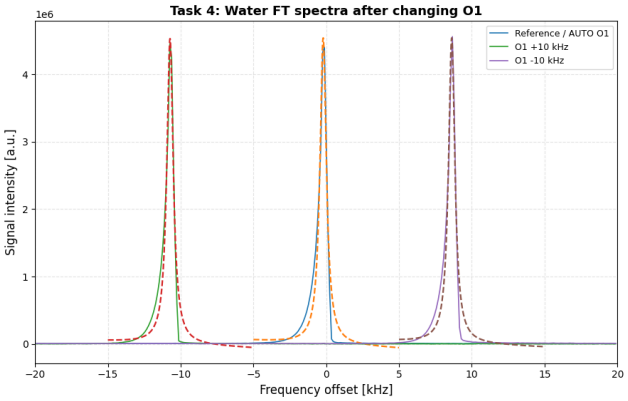


Figure 5: Frequency-domain peak shifts after changing the offset parameter $O1$

Spectrometer parameters: The spectrometer frequency used for the chemical-shift conversion was taken from the acquisition parameter file as

$$SF = 23{,}400{,}000 \text{ Hz} = 23.4000 \text{ MHz}. \tag{18}$$

The line labelled `Frequency` in the parameter file corresponds to the transmitter frequency including the offset contribution, whereas the parameter `SF` is the value used for converting frequency differences into ppm. A frequency offset of 10 kHz therefore corresponds to

$$\Delta\delta = \frac{10{,}000 \text{ Hz}}{23{,}400{,}000 \text{ Hz}} \times 10^6 = 427.35 \text{ ppm}. \tag{19}$$

Lorentzian fit results: All three Fourier-transformed spectra were fitted using a Lorentzian line shape with a linear baseline,

$$L(\nu) = c + m\nu + \frac{A\Gamma^2}{(\nu - \nu_0)^2 + \Gamma^2}, \tag{20}$$

where ν_0 is the fitted peak position, Γ is the half width at half maximum, and therefore 2Γ corresponds to the full width at half maximum.

Offset setting	ν_0 [kHz]	Fit unc. [kHz]	FWHM [kHz]
Ref. / AUTO O1	−0.212	0.009	0.531
O1 +10 kHz	−10.731	0.009	0.531
O1 −10 kHz	+8.618	0.008	0.528

Table 3: Lorentzian fit results for the water peak after changing the offset frequency O1.

Peak position comparison: The expected peak positions are determined from the applied offset change. Since increasing O1 shifts the observed peak in the opposite direction on the frequency-offset axis, the expected positions are 0 kHz for the reference measurement, −10 kHz for O1 +10 kHz, and +10 kHz for O1 −10 kHz.

The error in ppm was calculated from

$$\Delta\delta_{\mathrm{error}} = \frac{\Delta f}{SF} \times 10^6. \tag{21}$$

Peak [kHz]	Expected [kHz]	Abs. error [kHz]	Error [ppm]
−0.21	0.00	0.21	9.05
−10.73	−10.00	0.73	31.23
+8.62	+10.00	1.38	59.05

Table 4: Comparison of obtained peak frequencies with expected peak positions.

For example, for the largest deviation,

$$\Delta\delta_{\max} = \frac{1.38\ \text{kHz}}{23{,}400\ \text{kHz}} \times 10^6 = 59.05\ \text{ppm}. \tag{22}$$

The mean absolute deviation and the root-mean-square deviation are

$$\overline{|\Delta\delta|} = 33.11\ \text{ppm}, \tag{23}$$

$$\Delta\delta_{\text{RMS}} = 38.92\ \text{ppm}. \tag{24}$$

Thus, a reasonable estimate for the chemical-shift uncertainty from this measurement is

$$\boxed{\sigma_\delta \approx 39\ \text{ppm}} \tag{25}$$

or, using the most conservative estimate,

$$\boxed{\sigma_{\delta,\max} \approx 59\ \text{ppm}.} \tag{26}$$

Comparison with typical chemical shift ranges:

Nucleus	Shift range [ppm]	RMS uncertainty
^{1}H	0–15	259%
^{13}C	0–220	17.7%

Table 5: Comparison of the estimated RMS frequency-axis uncertainty with typical chemical shift ranges. The ^{13}C range is included only for general comparison; the present experiment was performed with ^{1}H NMR.

4.4.4 Discussion

Changing the offset frequency O1 shifts the position of the observed NMR peak on the frequency-offset axis. In the time domain, this corresponds to a change in the oscillation frequency of the free induction decay relative to the receiver reference frequency. If the transmitter frequency is exactly matched to the resonance frequency, the FID appears slowly varying after demodulation. If O1 is shifted, the signal acquires a faster oscillatory component. In the frequency domain, this appears as a displacement of the peak position.

The measured shifts followed the expected qualitative behaviour. Increasing O1 by 10 kHz shifted the peak to approximately −10 kHz, while decreasing O1 by 10 kHz shifted the peak to approximately +10 kHz. The fitted peak positions were therefore consistent with the expected direction of the frequency shift.

Quantitatively, however, the agreement was not exact. The reference peak was found at −0.21 kHz instead of 0 kHz, the O1 +10 kHz peak was found at −10.73 kHz instead of −10.00 kHz, and the O1 −10 kHz peak was found at +8.62 kHz instead of +10.00 kHz. The corresponding deviations were 9.05 ppm, 31.23 ppm, and 59.05 ppm, respectively.

These deviations are large when compared with ordinary ^{1}H chemical shift differences. Typical proton chemical shifts span only about 0–15 ppm, and many chemically relevant peak separations are much smaller than that. Therefore, the uncertainty estimated from this low-field experiment is not negligible for accurate chemical-shift determination. This is consistent with the limited spectral resolution of the minispec system, which is designed primarily for basic pulsed NMR measurements such as relaxation and diffusion experiments rather than high-resolution chemical shift spectroscopy.

The fact that the errors are not symmetric also suggests that the deviations are not purely random. Possible causes include imperfect frequency-axis calibration, residual phase or baseline errors after Fourier transformation, limited magnetic-field homogeneity, and the finite linewidth of the observed peak. The fitted FWHM values of approximately 0.53 kHz already correspond to about 23 ppm at this spectrometer frequency, meaning that the line itself is broad on the ppm scale.

In conclusion, the experiment correctly demonstrates how changing O1 shifts the NMR peak in the frequency domain. However, the quantitative peak-position accuracy is limited. The estimated RMS uncertainty of approximately 39 ppm, or up to 59 ppm using the largest observed deviation, is much larger than typical ^{1}H chemical-shift differences. Therefore, this setup is suitable for demonstrating the principle of frequency offsets and chemical-shift calibration, but it is not suitable for precise high-resolution chemical-shift measurements.

4.5 Hahn Echo Measurements

4.5.1 Theoretical Background

In this experiment, a Hahn echo pulse sequence is used to measure spin-echo signal attenuation as a function of echo time (2τ) for two samples; the resulting decay curves are fitted with a combined T_2-relaxation and diffusion model,

$$S = A \exp\left(-\frac{2\tau}{T_2}\right) \exp\left(-K\tau^3\right),$$

where A, T_2, and K are free fitting parameters to extract the transverse relaxation time T_2 and the diffusion-related constant K.

4.5.2 Measurement Procedure

The Hahn echo pulse sequence was loaded and the delay between the $\pi/2$ and π pulses was set to $\tau = 5000\ \mu$s. All acquisition parameters (O1, RD, and RG) were adjusted for the water sample and Sample #5 separately. The spin-echo signal intensity was recorded using the `MAG` and `SIZE` commands. The measurement was repeated for approximately 10 different τ values until the signal intensity dropped to

approximately 200 a.u. The collected τ values and corresponding signal amplitudes we then fitted with the combined transverse-relaxation and diffusion model.

4.5.3 Results

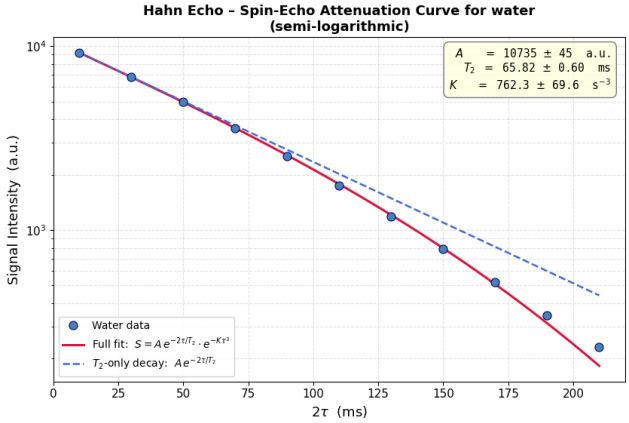


Figure 6: Hahn-echo attenuation curve for water.

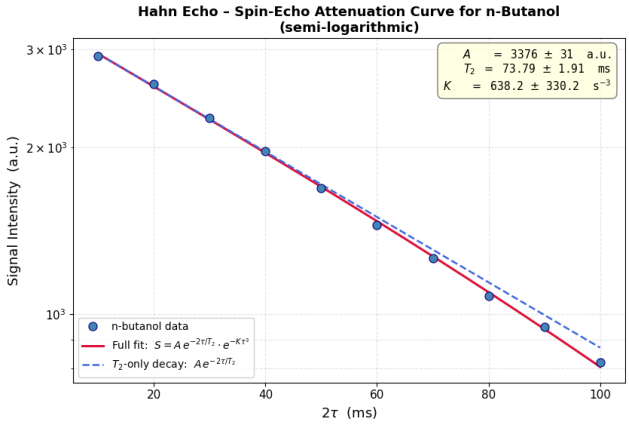


Figure 7: Hahn-echo attenuation curve for n-butanol.

Water:

$$A = 10735.4 \pm 45.1 \text{ a.u.}$$
$$T_2 = 65.82 \pm 0.60 \text{ ms}$$
$$K = 762.30 \pm 69.57 \text{ s}^{-3}$$

covariance matrix:

$$\begin{pmatrix} 2.03119380 \times 10^{3} & -2.16114777 \times 10^{-2} & -1.61365194 \times 10^{3} \\ -2.16114777 \times 10^{-2} & 3.58084336 \times 10^{-7} & 3.25854011 \times 10^{-2} \\ -1.61365194 \times 10^{3} & 3.25854011 \times 10^{-2} & 4.83948423 \times 10^{3} \end{pmatrix}$$

n-Butanol:

$$A = 3376.2 \pm 30.5 \text{ a.u.}$$
$$T_2 = 73.79 \pm 1.91 \text{ ms}$$
$$K = 638.18 \pm 330.18 \text{ s}^{-3}$$

covariance matrix:

$$\begin{pmatrix} 9.31809599 \times 10^{2} & -5.13427154 \times 10^{-2} & -6.86199617 \times 10^{3} \\ -5.13427154 \times 10^{-2} & 3.65824537 \times 10^{-6} & 5.63360686 \times 10^{-1} \\ -6.86199617 \times 10^{3} & 5.63360686 \times 10^{-1} & 1.09018040 \times 10^{5} \end{pmatrix}$$

4.5.4 Discussion

The spin-echo attenuation curves for both samples were fitted using

$$S(2\tau) = A \exp\left(-\frac{2\tau}{T_2}\right) \exp\left(-K\tau^3\right), \tag{27}$$

where the first term describes transverse relaxation and the second accounts for diffusion effects, with $K = \frac{2}{3}\gamma^2 D G^2$.

For the water sample, the fit yielded $T_2 = 65.82 \pm 0.60$ ms. The deviation between the full fit and the pure T_2 decay becomes clearly visible at larger τ, indicating that diffusion contributes significantly to the signal decay. The fitted diffusion parameter is $K = 762.3 \pm 69.6\ \text{s}^{-3}$.

For the n-butanol sample, the transverse relaxation time was found to be $T_2 = 73.79 \pm 1.91$ ms. In this case, the deviation from the pure T_2 decay is less pronounced, showing that diffusion has a weaker influence on the signal attenuation. This is also reflected in the smaller value of $K = 638.2 \pm 330.2\ \text{s}^{-3}$, although with a larger uncertainty.

Overall, the results show that the diffusion contribution is stronger in water, while n-butanol is dominated more by the exponential T_2 decay. The extracted T_2 values are of the same order of magnitude, with n-butanol showing a slightly longer relaxation time.

4.6 CPMG Measurements

4.6.1 Theoretical Background

Here the transverse relaxation behaviour by means of the Carr–Purcell–Meiboom–Gill (CPMG) pulse sequence is discussed. In contrast to the free induction decay, which is strongly influenced by magnetic-field inhomogeneity, and in contrast to the Hahn echo, which provides only a single refocused signal for a given pulse spacing, the CPMG experiment generates a whole train of echoes. The decay of these echoes can then be used to determine the transverse relaxation time T_2 more reliably.

The need for such a sequence arises from the fact that the transverse magnetization after a single 90° pulse decays

not only because of intrinsic spin-spin relaxation, but also because of reversible dephasing caused by static field inhomogeneities. The older manual explains that this effect broadens the line and masks the true transverse relaxation process. A Hahn echo can refocus part of this dephasing, but its decay still contains additional contributions, in particular diffusion effects that become more pronounced at larger pulse spacing τ. The CPMG sequence is introduced precisely to reduce this problem.

In the CPMG experiment, an initial 90° pulse creates transverse magnetization. This is followed by a series of 180° pulses separated by equal intervals. In phase-sensitive form, the sequence is written as

$$\left(\frac{\pi}{2}\right)_x - \tau - \left(\pi_y - 2\tau\right)_n . \tag{28}$$

Each 180° pulse reverses the phase evolution of the dephased spins and produces a new echo. In this way, the experiment can be understood as a repeated Hahn-echo sequence, in which the echo amplitudes are monitored over a longer total time interval.

According to the theoretical manual, the amplitude of the n-th echo is given by

$$M(2n\tau) = M(0)\exp\left(-\frac{2n\tau}{T_2}\right)\exp\left(-\frac{2}{3}\gamma^2 G^2 D\, n\tau^3\right), \tag{29}$$

where γ is the gyromagnetic ratio, G is the magnetic-field gradient, and D is the diffusion coefficient. If the total evolution time is written as $t = 2n\tau$, this expression becomes

$$M(t) = M(0)\exp\left(-\frac{t}{T_2}\right)\exp\left(-\frac{1}{3}\gamma^2 G^2 D\, t\tau^2\right). \tag{30}$$

The first exponential term describes the intrinsic transverse relaxation, while the second term accounts for additional decay caused by diffusion in the presence of a field gradient. This second contribution becomes smaller for sufficiently short values of τ. As a result, the CPMG sequence can approach the true transverse relaxation time more closely than a single Hahn-echo measurement.

A further advantage of the CPMG method over the original Carr–Purcell sequence lies in the phase shift of the refocusing pulses. In the original sequence, small errors in the 180° pulses accumulate from echo to echo and distort the measured amplitudes. The Meiboom–Gill modification avoids this by applying the refocusing pulses with a 90° phase shift relative to the initial 90° pulse. As a consequence, pulse imperfections are compensated much more effectively, and the even-numbered echoes in particular reproduce the true decay more reliably.

Therefore the goal here is to use a train of repeatedly refocused echoes to measure the decay of transverse magnetization under conditions in which reversible dephasing is strongly reduced. The resulting decay curve provides a practical route to the determination of T_2, while the comparison of measurements at different values of τ makes it possible to assess how strongly the measured decay is still affected by diffusion and pulse-spacing effects.

4.6.2 Measurement Procedure

For the CPMG measurements, the pulse sequence `cpmg.exe` was loaded in the acquisition mode of RiNMR. Before starting the measurement, the spectrometer was checked for proper tuning, since the practical guide notes that the resonance offset can drift over the course of the experiment. If necessary, the offset parameter was readjusted so that the signal remained close to resonance. [2]

Following the updated guide, the basic CPMG parameters were set to $SI = 1$, $NECH = 200$ while the echo-spacing parameter TAU was chosen in the range from 500 µs to 5000 µs. Here, $NECH$ specifies the number of recorded points or echoes in the decay trace, whereas TAU determines the spacing between the refocusing pulses. In addition to these settings, the remaining acquisition parameters such as RG, NS, DS, and RD were adjusted appropriately for the respective sample. [2]

The values of TAU and $NECH$ were then tuned empirically until the recorded signal showed a sufficiently clear decay over the observed time interval. In practice, this meant choosing the parameters such that the signal started at a high value and decayed substantially during the measurement window, while still remaining smooth enough for the subsequent fit. The aim was therefore not merely to observe the presence of echoes, but to record a complete decay curve that could be analyzed quantitatively.

After the acquisition parameters had been set, the measurement was carried out for two samples, namely one water sample and sample #5. According to the sample list of the older manual, sample #5 corresponds to n-butanol. For one of the investigated samples, the CPMG measurement was repeated with a second value of TAU in order to assess the influence of the echo spacing on the extracted relaxation behaviour. It is recommended to perform this comparison, for example, by doubling TAU and halving $NECH$ so that a similar total time interval is still covered. [2]

Once the measurement had been completed, the internal RiNMR analysis routine was executed by entering `T2` in the command line. The software then fitted the recorded decay and returned the corresponding time constant. The resulting fit plots were saved for later evaluation and comparison between the samples and between the different values of TAU.

4.6.3 Results

The CPMG measurements produced well-defined echo-train decay curves for both investigated samples. In each case, the signal decreased approximately exponentially with increasing time as shown in fig. 8, and the RiNMR fitting routine yielded a corresponding transverse relaxation time constant. The extracted results are summarized in Table 6.

Table 6: Fitted transverse relaxation times obtained from the CPMG measurements.

Sample	TAU (μs)	$NECH$	T_2 (ms)
Water	1000	200	70.820
Water	2000	100	70.568
Sample #5	10000	200	547.135

For the water sample, two CPMG measurements were performed with different echo spacings. For the first measurement, using $TAU = 1000\,\mu s$ and $NECH = 200$,the fit yielded $T_2 = 70.820\,ms$. For the second measurement, using $TAU = 2000\,\mu s$, $NECH = 100$, the resulting fit gave $T_2 = 70.568\,ms$. Thus, the two measurements for the water sample agree extremely well. The difference between the fitted values is only about 0.252 ms, which is small compared with the absolute value of the relaxation time. Within the precision of the present experiment, the extracted T_2 of the water sample can therefore be regarded as essentially independent of the chosen echo spacing in this parameter range.

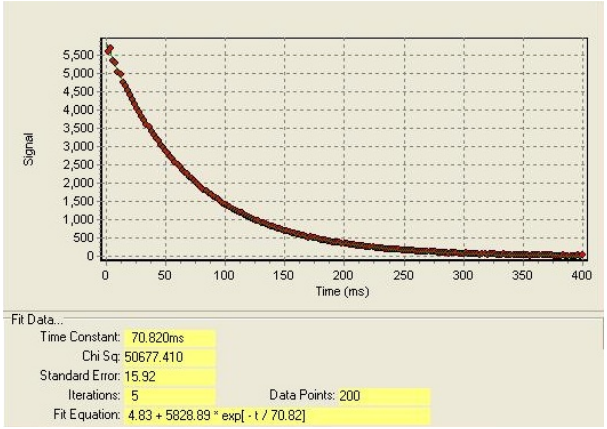
For sample #5, the CPMG measurement was carried out with $TAU = 10\,000\,\mu s$, $NECH = 200$, and the RiNMR fit yielded $T_2 = 547.135\,ms$. This value is substantially larger than the transverse relaxation time measured for the water sample. In other words, the echo-train signal of sample #5 decayed much more slowly and remained observable over a much longer time window.

4.6.4 Discussion

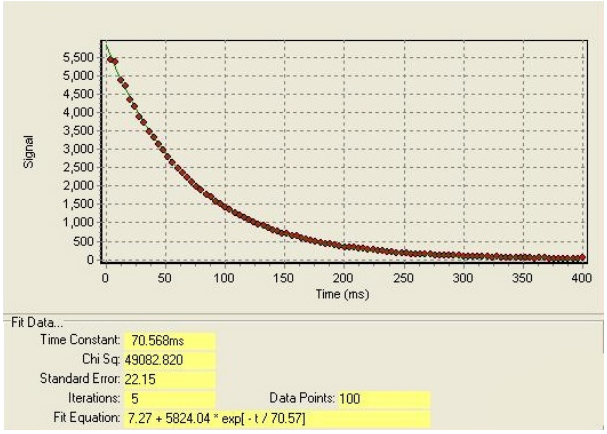
The CPMG measurements yielded internally consistent results and illustrate well the main purpose of the sequence, namely the determination of the transverse relaxation behaviour from an echo train rather than from a single echo. For the water sample the fitted relaxation times were $T_2 = 70.820\,ms$ and $T_2 = 70.568\,ms$. The difference between these two values is only about 0.252 ms, which is very small compared with the absolute magnitude of the relaxation time. This shows that, for the water sample, the extracted T_2 value is highly reproducible and only weakly dependent on the chosen echo spacing in this range.

This agreement is physically significant. In the theoretical description of the CPMG sequence, the decay contains not only the intrinsic transverse relaxation term but also an additional contribution that depends on the pulse spacing τ. If τ is chosen sufficiently small, this additional contribution becomes less important, and the measured decay approaches the true T_2. The fact that doubling TAU from 1000 µs to 2000 µs changed the fitted result only minimally indicates that, for the water sample, the measured decay is already dominated by the intrinsic relaxation rather than by a strong τ-dependent correction.

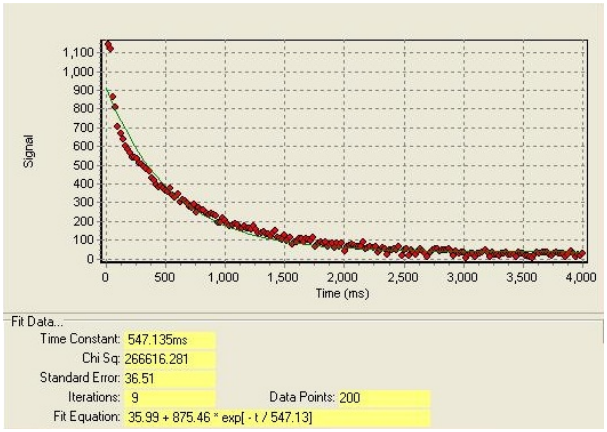
A clear contrast is observed for sample #5, for which the fit yielded $T_2 = 547.135\,ms$. This value is much larger than that of the water sample, which means that the transverse magne-



(a) Water sample, $TAU = 1000\,\mu s$, $NECH = 200$.



(b) Water sample, $TAU = 2000\,\mu s$, $NECH = 100$.



(c) Sample #5, $TAU = 10\,000\,\mu s$, $NECH = 200$.

Figure 8: CPMG decay curves and corresponding RiNMR fits for the investigated samples. The two upper panels show the repeated measurements for the water sample at different echo spacings, while the lower panel shows the result for sample #5.

tization in sample #5 remains coherent over a much longer time interval. Experimentally, this is directly visible in the much slower decay of the recorded echo train. The difference between the two samples is therefore not marginal but substantial, with sample #5 showing a transverse relaxation time that is roughly an order of magnitude larger than that of water.

At the same time, the interpretation of the sample #5 result requires some caution. In this case, the measurement was performed with $TAU = 10\,000\,\mu s$, which is larger than the range suggested in the practical guide. Since the theoretical expression for the CPMG decay contains a τ-dependent correction term, a larger pulse spacing may enhance additional decay contributions related to diffusion or imperfect refocusing. For this reason, the fitted value for sample #5 should be regarded as the transverse relaxation time obtained under the chosen experimental conditions, but not necessarily as a value that is as robust against pulse-spacing effects as the water result. A second CPMG measurement for sample #5 at a smaller value of TAU would have made this point easier to assess quantitatively.

Another important aspect is the role of pulse accuracy. The CPMG sequence is intended to reduce the accumulation of pulse errors through the Meiboom–Gill phase modification. The good agreement between the two water measurements suggests that the pulse calibration from section 4.1 was sufficiently accurate for the sequence to work reliably. Had the 180° pulses been significantly misadjusted, a stronger dependence of the fitted decay on the measurement parameters would have been expected.

5. Conclusion

Overall, the experiment successfully demonstrated the central principles of pulsed ^{1}H NMR in liquids. The pulse calibration gave $P90 = 2.85\,\mu s$ and $P180 = 5.70\,\mu s$, consistent with the expected factor-of-two relation. The inversion-recovery measurements produced reasonable sample-dependent longitudinal relaxation times, with water showing the longest value, $T_1 = 332.522\,ms$, and sample #1 the shortest value, $T_1 = 6.035\,ms$. Oil and sample #5 showed intermediate values of $T_1 = 51.720\,ms$ and $T_1 = 106.064\,ms$, respectively.

The FID and Fourier-transform analysis connected the time-domain decay to the frequency-domain linewidth, giving an effective transverse decay time of $T_2 \approx 601\,\mu s$. The chemical-shift offset task showed the expected qualitative peak shifts when $O1$ was changed, but the estimated uncertainty of approximately 38.92 ppm RMS indicates that this low-field setup is not suitable for precise high-resolution chemical-shift determination.

The echo-based measurements gave transverse relaxation times of $T_2 = (65.82 \pm 0.60)\,ms$ for water and $T_2 = (73.79 \pm 1.91)\,ms$ for n-butanol from Hahn echo. The CPMG measurements gave reproducible water values of $T_2 = 70.820\,ms$ and $T_2 = 70.568\,ms$, while sample #5 gave a much larger value of $T_2 = 547.135\,ms$, which should be interpreted cautiously because of the large TAU used. The main limitations were pulse imperfections, resonance drift, RF-field inhomogeneity, receiver dead time, signal noise, fitting-window choices, and diffusion-related attenuation. Despite these limitations, the measured values are mostly physically meaningful and demonstrate the expected behaviour of pulsed NMR relaxation experiments. [1, 2]

References

[1] *Nuclear Magnetic Resonance (NMR) in Liquids. Experiment for the Advanced Physics Labs*. Laboratory manual; Supervisor: Dr. R. Valiullin. University of Leipzig, Department of Physics and Geosciences. 2009.

[2] *A Short Guide to NMR Experiments. MARAN Ultra – RiNMR*. Undated practical guide for RiNMR measurements. University of Leipzig.

Appendix

Summary of Relevant Experimental Values

Table 7: Summary of the main numerical results obtained throughout the experiment.

Experiment / Quantity	Value	Comment
Pulse calibration		
90° pulse length	$P90 = 2.85\,\mu s$	Automatically determined by `Auto P90`
180° pulse length	$P180 = 5.70\,\mu s$	Consistent with $P180 = 2P90$
Inversion recovery: longitudinal relaxation		
Water	$T_1 = 332.522\,ms$	Longest measured T_1 value
Water	$\tau_0 = 200.697\,ms$	Null point from RiNMR fit
Sample #1	$T_1 = 6.035\,ms$	Shortest measured T_1 value
Sample #1	$\tau_0 = 4.046\,ms$	Null point from RiNMR fit
Sample #5	$T_1 = 106.064\,ms$	Intermediate relaxation time
Sample #5	$\tau_0 = 93.816\,ms$	Null point from RiNMR fit
Oil	$T_1 = 51.720\,ms$	Intermediate relaxation time
Oil	$\tau_0 = 30.068\,ms$	Null point from RiNMR fit
FID and Fourier-transform analysis		
Spectrometer frequency	$SF = 23.7006\,MHz$	Extracted from parameter file
Real-channel maximum	≈ 6700 a.u.	Approximate maximum from time-domain FID
Imaginary-channel maximum	≈ 3600 a.u.	Approximate maximum from time-domain FID
FID decay window	$\approx 8\,ms$	Signal decays close to baseline within this time
Peak position	$\nu_0 = -0.000\,17\,MHz$	Lorentzian peak position
Half-width at half maximum	$\gamma = 0.000\,26\,MHz$	Lorentzian half-width parameter
Full width at half maximum	$FWHM = 0.000\,530\,MHz$	Equivalent to approximately 529.7 Hz
Effective transverse decay time	$T_2^* \approx 601\,\mu s$	Derived from linewidth; includes inhomogeneous broadening
Lorentzian amplitude	$A \approx 4.5445 \times 10^6$ a.u.	Fit amplitude
Fit offset	$c = 15274$ a.u.	Baseline offset from fit
Fit quality	$R^2 = 0.9682$	Lorentzian fit quality
Chemical-shift / offset analysis		
Spectrometer frequency for offset conversion	$SF = 23.4000\,MHz$	Used for frequency-to-ppm conversion in offset task
Applied offset step	$\Delta f = 10\,kHz$	Applied change in $O1$
Equivalent shift of 10 kHz	$\Delta\delta = 427.35\,ppm$	Frequency-to-ppm conversion result
Reference peak position	$\nu_0 = -0.212\,kHz$	From Lorentzian fit
O1 +10 kHz peak position	$\nu_0 = -10.731\,kHz$	Expected near −10 kHz
O1 −10 kHz peak position	$\nu_0 = 8.618\,kHz$	Expected near +10 kHz
Reference absolute error	0.21 kHz, 9.05 ppm	Error relative to expected peak position
O1 +10 kHz absolute error	0.73 kHz, 31.23 ppm	Error relative to expected peak position
O1 −10 kHz absolute error	1.38 kHz, 59.05 ppm	Largest observed deviation
Mean absolute deviation	$\overline{\lvert\Delta\delta\rvert} = 33.11\,ppm$	Mean absolute frequency-axis error
RMS deviation	$\Delta\delta_{RMS} = 38.92\,ppm$	Main uncertainty estimate
Maximum deviation	$\Delta\delta_{max} = 59.05\,ppm$	Conservative uncertainty estimate
Hahn echo measurements		
Water amplitude	$A = 10735.4 \pm 45.1$ a.u.	Initial amplitude parameter
Water transverse relaxation	$T_2 = (65.82 \pm 0.60)\,ms$	Hahn-echo fit result
Water diffusion parameter	$K = (762.30 \pm 69.57)\,s^{-3}$	Diffusion-related fit parameter
n-butanol amplitude	$A = 3376.2 \pm 30.5$ a.u.	Initial amplitude parameter
n-butanol transverse relaxation	$T_2 = (73.79 \pm 1.91)\,ms$	Hahn-echo fit result

Experiment / Quantity	Value	Comment
n-butanol diffusion parameter	$K = (638.18 \pm 330.18)\ \mathrm{s}^{-3}$	Diffusion-related fit parameter
CPMG measurements		
Water CPMG, first run	$TAU = 1000\,\mu\mathrm{s},\ NECH = 200$	Measurement settings
Water CPMG, first run	$T_2 = 70.820\,\mathrm{ms}$	Fitted transverse relaxation time
Water CPMG, second run	$TAU = 2000\,\mu\mathrm{s},\ NECH = 100$	Measurement settings
Water CPMG, second run	$T_2 = 70.568\,\mathrm{ms}$	Fitted transverse relaxation time
Sample #5 CPMG	$TAU = 10\,000\,\mu\mathrm{s},\ NECH = 200$	Larger than recommended TAU range
Sample #5 CPMG	$T_2 = 547.135\,\mathrm{ms}$	Much longer than water; should be interpreted cautiously
Useful comparisons		
Typical ^{1}H shift range	0–15 ppm	Used for comparison with offset uncertainty
Typical ^{13}C shift range	0–220 ppm	Included only for general comparison
RMS uncertainty relative to ^{1}H range	259%	Shows poor high-resolution chemical-shift accuracy
RMS uncertainty relative to ^{13}C range	17.7%	Still significant compared with full ^{13}C range

Chemical-Shift Offset Analysis

Table 8: Lorentzian fit results for the water peak after changing the offset frequency O1.

Offset setting	Peak position ν_0 [kHz]	Fit uncertainty [kHz]	FWHM [kHz]
Reference / AUTO O1	−0.212	0.009	0.531
O1 +10 kHz	−10.731	0.009	0.531
O1 −10 kHz	+8.618	0.008	0.528

Table 9: Comparison between measured and expected peak positions in the O1-offset experiment.

Measurement	Expected [kHz]	Absolute error [kHz]	Error [ppm]
Reference / AUTO O1	0.00	0.21	9.05
O1 +10 kHz	−10.00	0.73	31.23
O1 −10 kHz	+10.00	1.38	59.05

Table 10: Comparison of the estimated RMS frequency-axis uncertainty with typical chemical-shift ranges.

Nucleus	Typical chemical-shift range [ppm]	RMS uncertainty as percentage of range
^{1}H	0–15	259%
^{13}C	0–220	17.7%

In plain words: **the Hahn echo value is probably a lower, more diffusion-/fit-affected estimate, while the CPMG value is a longer apparent decay time from repeated refocusing, but the sample #5 CPMG setting was non-ideal and needs caution.** Your own report also notes that the Hahn echo fit included a diffusion-related term and gave $T_2 = (73.79 \pm 1.91)$ ms for n-butanol, while the CPMG measurement for sample #5 used $TAU = 10\,000\,\mu\mathrm{s}$ and returned $T_2 = 547.135$ ms